
QED NOTES

Pavel Okun

Department of Chemistry, University of Michigan, Ann Arbor USA

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Part I

Dirac Field

1. PAULI & DIRAC MATRICES

I use the same convention in these notes as Schwartz and Peskin & Schroeder. The Pauli matrices form a 4-vector (I use arrows for 4-vectors and bold-face for 3-vectors)

$$\vec{\sigma} = (\sigma^0, \sigma^1, \sigma^2, \sigma^3) = (I_2, \boldsymbol{\sigma}), \quad (1)$$

where

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (2)$$

and I_n is the n -dimensional identity matrix

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (3)$$

I define the conjugate Pauli vector

$$\bar{\boldsymbol{\sigma}} = -\boldsymbol{\sigma}, \quad \vec{\bar{\sigma}} = (I_2, -\boldsymbol{\sigma}). \quad (4)$$

Some properties are

$$(\sigma^i)^2 = I_4, \quad \sigma^{i\dagger} = \sigma^i. \quad (5)$$

The Dirac matrices in the Weyl Representation are

$$\gamma^0 = \begin{pmatrix} 0_2 & I_2 \\ I_2 & 0_2 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0_2 & \sigma^i \\ -\sigma^i & 0_2 \end{pmatrix} \quad (6)$$

Each element is a 2×2 matrix, so the Dirac matrices have dimension 4×4 . Thus 0_n is the n -dimensional zero matrix.

$$0_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad (7)$$

The gamma matrices obey

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4, \quad (8)$$

where η is the Minkowski metric $\eta^{\alpha\beta} = \text{diag}(1, -1, -1, -1)$. This property implies

$$(\gamma^0)^2 = I_4, \quad (\gamma^a)^2 = -I_4. \quad (9)$$

Other useful properties are

$$\gamma^0 \gamma^{\mu\dagger} \gamma^0 = \gamma^\mu, \quad \gamma^{i\dagger} = -\gamma^i. \quad (10)$$

It is useful to define another Dirac matrix

$$\gamma^5 = \begin{pmatrix} -I_2 & 0_2 \\ 0_2 & I_2 \end{pmatrix}. \quad (11)$$

Its properties are

$$(\gamma^5)^2 = 1, \quad \{\gamma^5, \gamma^\mu\} = 0. \quad (12)$$

2. LORENTZ GROUP

The Lorentz transformations that act on 4-vectors in SR, $x^{\bar{\mu}} = \Lambda^{\bar{\mu}}_{\mu} x^{\mu}$ are one representation of the Lorentz group. In this representation the rotation $\mathbf{J}$ and boost $\mathbf{K}$ generators are

$$\begin{aligned} J_1 &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & J_2 &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \\ J_3 &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} & K_1 &= i \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ K_2 &= i \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} & K_3 &= i \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (13)$$

Any Lorentz transformation can be written from these generators

$$\Lambda = \exp(i\theta^a J_a + i\beta^b K_b), \quad (14)$$

where the θ_i are rotation angles and the β_i are boost angles (rapidities).

The generators obey the algebra

$$[J_i, J_j] = i\epsilon_{ijk} J_k, \quad [J_i, K_j] = i\epsilon_{ijk} K_k, \quad [K_i, K_j] = -i\epsilon_{ijk} J_k, \quad (15)$$

where there is no sum over k and ϵ_{ijk} is the Levi-Civita symbol.

2.1. Irreducible Representations

I can break the Lorentz group into 2 3D rotation groups with generators

$$\mathbf{J}^{\pm} = \frac{\mathbf{J} \pm i\mathbf{K}}{2}, \quad (16)$$

that satisfy the algebra

$$[J_i^{\pm}, J_j^{\pm}] = i\epsilon_{ijk} J_k^{\pm}, \quad [J_i^+, J_j^-] = 0. \quad (17)$$

Using this irreducible representation I can generate all the representations of the Lorentz group which are characterized by the total angular momentum quantum numbers (j^+, j^-) .

3. SPINORS

There are two types (left handed and right handed) Weyl spinor fields

$$\psi_R(\vec{x}) = \begin{pmatrix} \psi_R^1(\vec{x}) \\ \psi_R^2(\vec{x}) \end{pmatrix} \quad \psi_L(\vec{x}) = \begin{pmatrix} \psi_L^1(\vec{x}) \\ \psi_L^2(\vec{x}) \end{pmatrix}. \quad (18)$$

We see that Weyl spinor fields have two components. What's the difference between the left and right handed spinors? Left handed spinors transform under the (1/2,0) irreducible Lorentz representation (Sec. 2.1) with rotation and boost generators

$$\mathbf{J} = \frac{\boldsymbol{\sigma}}{2}, \quad \mathbf{K} = \frac{i}{2}\boldsymbol{\sigma}. \quad (19)$$

Right handed spinors transform under (0,1/2) and have generators

$$\mathbf{J} = \frac{\boldsymbol{\sigma}}{2}, \quad \mathbf{K} = -\frac{i}{2}\boldsymbol{\sigma}. \quad (20)$$

The electron field is a 4-component spinor made from a left-handed and right-handed Weyl spinor

$$\psi(\vec{x}) = \begin{pmatrix} \psi_L(\vec{x}) \\ \psi_R(\vec{x}) \end{pmatrix} = \begin{pmatrix} \psi_1(\vec{x}) \\ \psi_2(\vec{x}) \\ \psi_3(\vec{x}) \\ \psi_4(\vec{x}) \end{pmatrix}. \quad (21)$$

This type of spinor is called a Dirac spinor. The Lorentz representation Dirac spinors transform under is the direct sum (1/2,0) $\oplus$ (0,1/2). Its generators are

$$S^{\alpha\beta} = \frac{i}{4}[\gamma^\alpha, \gamma^\beta], \quad (22)$$

where the rotation and boost generators are

$$J_1 = S^{23}, \quad J_2 = -S^{13}, \quad J_3 = S^{12}, \quad K_i = S^{0i}. \quad (23)$$

I confirmed that these generators satisfy the Lorentz algebra in Eq. 15.

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \quad [J_i, K_j] = i\epsilon_{ijk}K_k, \quad [K_i, K_j] = -i\epsilon_{ijk}J_k, \quad (24)$$

4. CFT

Right now we are doing classical physics and $\psi(\vec{x})$ is a classical field. It's free Lagrangian is

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi, \quad (25)$$

where $\not{\partial} = \gamma^\mu \partial_\mu$ and the conjugate Dirac spinor is

$$\bar{\psi} = \psi^\dagger \gamma^0 = (\psi_R^\dagger \quad \psi_L^\dagger). \quad (26)$$

The classical equation of motion of the free electron field is the famed Dirac equation

$$(i\not{\partial} - m)\psi(\vec{x}) = 0. \quad (27)$$

5. QUANTIZED DIRAC FIELD

The Dirac field ψ is a spinor field which describes spin-1/2 particles like the electron. It is time to quantize it. The Dirac Field and its conjugate are

$$\begin{aligned} \psi(\vec{x}) &= \sum_{s=1}^2 \int d^3p \frac{a_{\mathbf{p}}^s u_s(\vec{p}) e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^{s\dagger} v_s(\vec{p}) e^{i\vec{p}\cdot\vec{x}}}{\sqrt{2E_p} (2\pi)^3} \\ \bar{\psi}(\vec{x}) &= \sum_{s=1}^2 \int d^3p \frac{a_{\mathbf{p}}^{s\dagger} \bar{u}_s(\vec{p}) e^{i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^s \bar{v}_s(\vec{p}) e^{-i\vec{p}\cdot\vec{x}}}{\sqrt{2E_p} (2\pi)^3} \end{aligned} \quad (28)$$

There's a lot going on in these definitions. I will explain all of it. As in scalar field theory the energy is

$$E_p = \sqrt{p^2 + m^2}, \quad p = |\mathbf{p}|. \quad (29)$$

The a create and annihilate particles and the b create and annihilate antiparticles. The u and v objects are Dirac spinors (so they have 4 components)

$$u_s(\vec{p}) = \begin{pmatrix} \sqrt{\vec{p}\cdot\vec{\sigma}} \chi_s \\ \sqrt{\vec{p}\cdot\vec{\sigma}} \chi_s \end{pmatrix}, \quad v_s(\vec{p}) = \begin{pmatrix} \sqrt{\vec{p}\cdot\vec{\sigma}} \chi_s \\ -\sqrt{\vec{p}\cdot\vec{\sigma}} \chi_s \end{pmatrix}, \quad (30)$$

where the χ_s are two components spinors

$$\chi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (31)$$

I prefer to have write all 4-components explicitly, so I work them out.

$$\begin{aligned} \vec{p}\cdot\vec{\sigma} &= (E, \mathbf{p}) \cdot (I_2, \boldsymbol{\sigma}) = EI_2 - \mathbf{p}\cdot\boldsymbol{\sigma} \\ \vec{p}\cdot\vec{\bar{\sigma}} &= (E, \mathbf{p}) \cdot (I_2, -\boldsymbol{\sigma}) = EI_2 + \mathbf{p}\cdot\boldsymbol{\sigma} \end{aligned} \quad (32)$$

$$\mathbf{p}\cdot\boldsymbol{\sigma} = p_x\sigma_1 + p_y\sigma_2 + p_z\sigma_3 \quad (33)$$

$$\sqrt{\vec{p}\cdot\vec{\sigma}} = \frac{1}{2p} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad \sqrt{\vec{p}\cdot\vec{\bar{\sigma}}} = \frac{1}{2p} \begin{pmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{pmatrix}, \quad (34)$$

$$A_{11} = \sqrt{E-p}(p+p_z) + \sqrt{E+p}(p-p_z) \quad (35)$$

$$A_{12} = (p_x - ip_y) \left(\sqrt{E-p} - \sqrt{E+p} \right) \quad (36)$$

$$A_{21} = (p_x + ip_y) \left(\sqrt{E-p} - \sqrt{E+p} \right) \quad (37)$$

$$A_{22} = \sqrt{E+p}(p+p_z) + \sqrt{E-p}(p-p_z) \quad (38)$$

$$(39)$$

$$\begin{aligned} u_1(\vec{p}) &= \frac{1}{2p} \begin{pmatrix} A_{11} \\ A_{21} \\ A_{22} \\ -A_{21} \end{pmatrix}, & u_2(\vec{p}) &= \frac{1}{2p} \begin{pmatrix} A_{12} \\ A_{22} \\ -A_{12} \\ A_{11} \end{pmatrix} \\ v_1(\vec{p}) &= \frac{1}{2p} \begin{pmatrix} A_{11} \\ A_{21} \\ -A_{22} \\ A_{21} \end{pmatrix}, & v_2(\vec{p}) &= \frac{1}{2p} \begin{pmatrix} A_{12} \\ A_{22} \\ A_{12} \\ -A_{11} \end{pmatrix} \end{aligned} \quad (40)$$

I get the conjugates from Eq. 44.

$$\begin{aligned}\bar{u}_1(\vec{p}) &= \frac{(A_{22} \ -A_{12} \ A_{11} \ A_{12})}{2p} \\ \bar{u}_2(\vec{p}) &= \frac{(-A_{21} \ A_{11} \ A_{21} \ A_{22})}{2p} \\ \bar{v}_1(\vec{p}) &= \frac{(-A_{22} \ A_{12} \ A_{11} \ A_{12})}{2p} \\ \bar{v}_2(\vec{p}) &= \frac{(A_{21} \ -A_{11} \ A_{21} \ A_{22})}{2p}\end{aligned}\quad (41)$$

5.1. Inner & Outer Spinor Products

Dirac spinors are 4-component objects but to avoid clutter QED notation often suppresses the indices that refer to these components. The Dirac spinor is a column vector

$$(\psi)^\alpha = \begin{pmatrix} (\psi)^1 \\ (\psi)^2 \\ (\psi)^3 \\ (\psi)^4 \end{pmatrix}, \quad (42)$$

where I have introduced an index to track the component. From now on I will always put spinor component indices outside of parentheses to avoid confusion. The Dirac conjugate spinor is

$$\bar{\psi} = \psi^\dagger \gamma^0. \quad (43)$$

Doing this matrix product in Mathematica yields the row vector

$$\begin{aligned}(\bar{\psi})_\alpha &= ((\bar{\psi})_1 \ (\bar{\psi})_2 \ (\bar{\psi})_3 \ (\bar{\psi})_4) = \\ &((\psi)^{3*} \ (\psi)^{4*} \ (\psi)^{1*} \ (\psi)^{2*})\end{aligned}\quad (44)$$

I define the inner product between two Dirac spinors ψ_1 and ψ_2

$$\begin{aligned}\bar{\psi}_1 \psi_2 &= ((\psi_1)^{3*} \ (\psi_1)^{4*} \ (\psi_1)^{1*} \ (\psi_1)^{2*}) \begin{pmatrix} (\psi_2)^1 \\ (\psi_2)^2 \\ (\psi_2)^3 \\ (\psi_2)^4 \end{pmatrix} \\ &= (\psi_1)^{3*}(\psi_2)^1 + (\psi_1)^{4*}(\psi_2)^2 + (\psi_1)^{1*}(\psi_2)^3 + (\psi_1)^{2*}(\psi_2)^4 = \\ &(\bar{\psi}_1)_\mu (\psi_2)^\mu.\end{aligned}\quad (45)$$

It does not commute: $\bar{\psi}_1 \psi_2 \neq \bar{\psi}_2 \psi_1$. Using my explicit expressions in Eqs. 40 & 41 I confirmed the identities

$$\begin{aligned}\bar{u}_s(\vec{p}) u_{s'}(\vec{p}) &= 2m \delta_{ss'}, \quad \bar{v}_s(\vec{p}) v_{s'}(\vec{p}) = -2m \delta_{ss'}, \\ \bar{u} v &= \bar{v} u = 0.\end{aligned}\quad (46)$$

There is also an outer product $\psi_1 \bar{\psi}_2$ which yields a 4×4 matrix. In terms of components

$$(\psi_1 \bar{\psi}_2)_\beta^\alpha = (\psi_1)^\alpha (\bar{\psi}_2)_\beta. \quad (47)$$

There are two important outer product identities which I confirmed with my explicit expressions

$$\sum_{s=1}^2 u_s(\vec{p}) \bar{u}_s(\vec{p}) = \not{p} + m, \quad \sum_{s=1}^2 v_s(\vec{p}) \bar{v}_s(\vec{p}) = \not{p} - m. \quad (48)$$

I have suppressed the identity in $I_4 m$. You should know it's there because these are matrix equations.

6. FEYNMAN PROPAGATOR

The Dirac Field Feynman propagator is

$$\langle 0 | T \psi(0) \bar{\psi}(\vec{x}) | 0 \rangle = i \int \frac{d^4 p}{(2\pi)^4} \frac{\not{p} + m I_4}{\vec{p}^2 - m^2 + i\epsilon} e^{i\vec{p} \cdot \vec{x}}, \quad (49)$$

where $|0\rangle$ is the Dirac free field vacuum, T is the time ordering operator, and there is an implied $\epsilon \rightarrow 0_+$. I will leave the identity implicit to get a more aesthetic expression

$$\langle 0 | T \psi(0) \bar{\psi}(\vec{x}) | 0 \rangle = i \int \frac{d^4 p}{(2\pi)^4} \frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon} e^{i\vec{p} \cdot \vec{x}}. \quad (50)$$

Part II

Electromagnetic Field

The photon is a massless spin-1 particle which is described classically by the 4-vector potential $\vec{A}$, which is what I will quantize. It is related to the electromagnetic field tensor by

$$F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha. \quad (51)$$

The free Lagrangian is

$$4\mathcal{L} = F_{\mu\nu}F^{\nu\mu}. \quad (52)$$

The classical equations of motion are

$$\square A_\mu = \partial_\mu \partial_\nu A^\nu - \partial_\nu \partial_\mu A^\nu. \quad (53)$$

This field also has a gauge symmetry. The transform

$$A_\mu(\vec{x}) \rightarrow A_\mu(\vec{x}) + \partial_\mu f(\vec{x}), \quad (54)$$

for any function $f(\vec{x})$ does not change the physics. The field operator is

$$A_\mu(\vec{x}) = \sum_{j=1}^2 \int d^3p \frac{\epsilon_\mu^j(\vec{p}) a_{\vec{p}}^j e^{-i\vec{p}\cdot\vec{x}} + \epsilon_\mu^{j*}(\vec{p}) a_{\vec{p}}^{j\dagger} e^{i\vec{p}\cdot\vec{x}}}{(2\pi)^3 \sqrt{2E_p}}, \quad (55)$$

where $E_p = p^0$ is the on-shell energy, and the ϵ_μ^j are polarization vectors. There are two polarization vectors indexed by the spin indices $\vec{e}_j$. The μ above indexes the components. Electromagnetic waves are transverse polarized so these objects obey

$$\vec{p} \cdot \vec{e}_j = \vec{p}^2 = 0, \quad \vec{e}_i \cdot \vec{e}_j = -\delta_{ij}. \quad (56)$$

Notice that they are timelike and $\vec{p}$ is null (it is the 4-momentum of a photon). The polarization components can be complex. I will pick explicit polarization vectors appropriate for particular problems. The Feynman propagator for the photon is

$$\Pi^{\mu\nu}(\vec{p}) = \frac{1}{\vec{p}^2 + i\epsilon} \left(\frac{1-\xi}{\vec{p}^2} p^\mu p^\nu - \eta^{\mu\nu} \right), \quad (57)$$

which looks like it has a divergence (it doesn't) and where ξ parameterizes the gauge. I keep things simple and use the Feynman-'t Hooft gauge: $\xi = 1$ (because I am following Schwartz)

$$\Pi^{\mu\nu}(\vec{p}) = -\frac{\eta^{\mu\nu}}{\vec{p}^2 + i\epsilon}. \quad (58)$$

Part III

Interactions

7. LAGRANGIAN

The QED Lagrangian for the electromagnetic field and a charge $-e$ Dirac field, the electron, is

$$\mathcal{L} = -\frac{F^2}{4} - m\bar{\psi}\psi + i\bar{\psi}\not{D}\psi, \quad (59)$$

where $F^2 = F_{\alpha\beta}F^{\alpha\beta}$ and $\not{D} = \gamma^\mu D_\mu$ and D_μ is the covariant derivative

$$D_\mu = \partial_\mu + ieA_\mu. \quad (60)$$

Expanding out the covariant derivative in the Lagrangian yields

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 \quad (61)$$

where (see Eqs. 25 & 52) the free Lagrangian is

$$\mathcal{L}_0 = -\frac{F^2}{4} + \bar{\psi}(i\not{D} - m)\psi, \quad (62)$$

and the interaction Lagrangian has a single term.

$$\mathcal{L}_1 = -e\bar{\psi}\gamma^\mu\psi A_\mu. \quad (63)$$

8. FEYNMAN RULES

The Feynman rules for constructing diagrams are more complicated for QED than for Scalar field theory. I still integrate over internal momenta and conserve 4-momenta at each vertex. This section is a work in progress. I will add new rules as I need them.

Electron arrows point to the right, positron arrows point to the left. Electrons and positrons are straight lines and photons are wavy lines.

External fermion lines get spinors: u for electrons and v for positrons. Dirac conjugates (bars) distinguish incoming particles from outgoing particles. These rules are summarized in Fig. 1. External electrons/positrons are on-shell and free (the interaction happens in the interior of the diagram). The Dirac equation for free spinor fields yields (I derived this in HW 8)

$$\begin{aligned} (\not{p} - m)u_s(\vec{p}) &= \bar{u}_s(\vec{p})(\not{p} - m) = \\ (\not{p} + m)v_s(\vec{p}) &= \bar{v}_s(\vec{p})(\not{p} + m) = 0. \end{aligned} \quad (64)$$

I give the rule for assigning external photon polarizations in Fig. 2.

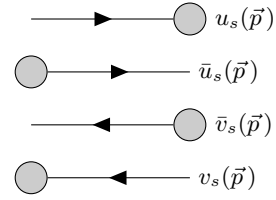


FIG. 1. Feynman rules for external electrons/positrons. The blobs represents the rest of the diagram.

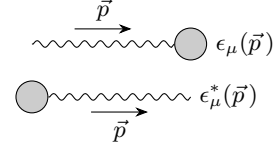


FIG. 2. Feynman rules for external photons.

Virtual (internal) photons contribute a photon propagator (Fig. 3). Like Schwartz I work in the Feynman gauge $\xi = 1$

$$\Pi_{\mu\nu}(\vec{p}) = -\frac{\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon}. \quad (65)$$

The electron propagator is

$$i\frac{\not{p} + m}{\vec{p}^2 - m^2 + i\epsilon}. \quad (66)$$

There is only one term (Fig. 4) in the interaction Lagrangian, so there is only one interaction vertex, but it has 4 variants. Each variant contributes the factor

$$-ie\gamma^\mu. \quad (67)$$

The γ matrix must always be sandwiched between two spinors like so $\bar{u}\gamma^\mu u$. Order matters if you suppress spinor component indices. The μ index must be contracted with the vertex photon propagator. Also note that $\bar{u}u$ is the order for the scalar product. Charge and momentum must be conserved at each vertex.

The relative sign of different diagrams matter. The overall sign does not since observables depend on $|\mathcal{M}|^2$, where $\mathcal{M}$ is the non-trivial part of the S-matrix. The only sign rule I need right now is to flip the sign when exchanging external identical fermions. QED diagrams also have symmetry factors that must be calculated by counting Wick theorem contractions. I describe the process in my Real Scalar Field Notes. Very often the QED symmetry factor is 1. But this is not comforting because it might not be 1 and you still have to check each diagram. I assign factors of $(2\pi)^4$ the same way as I do for scalar fields. Closed fermion loops are traced over and get a minus sign. Dirac matrices are multiplied against the direction of particle flow in a fermion line.

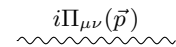


FIG. 3. Virtual photon.

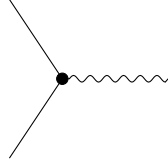
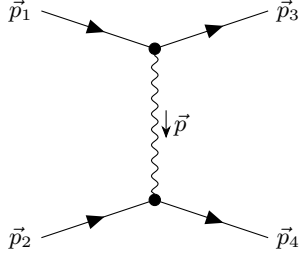


FIG. 4. Fundamental QED vertex: two fermions meet a photon.


 FIG. 5. Feynman diagram $\mathfrak{Y}$.

$$9. \quad e^- e^- \rightarrow e^- e^-$$

In this Sec. I follow Schwartz Sec. 13.1.1. The initial momenta are p_1 and p_2 and the final momenta are p_3 and p_4 . Because this is a 2 to 2 scattering I use Mandelstam variables

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = (\vec{p}_3 + \vec{p}_4)^2, \\ t &= (\vec{p}_1 - \vec{p}_3)^2 = (\vec{p}_2 - \vec{p}_4)^2, \\ u &= (\vec{p}_1 - \vec{p}_4)^2 = (\vec{p}_2 - \vec{p}_3)^2. \end{aligned} \quad (68)$$

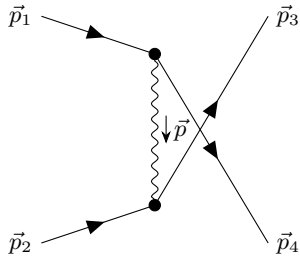
I define the total momenta

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{p}_3 + \vec{p}_4 \quad (69)$$

The matrix element to leading order is

$$i\delta^{(4)}(\vec{P} - \vec{K})\mathcal{M} = \mathfrak{Y} - \mathfrak{Y} \quad (70)$$

where the Feynman diagrams are in Figs. 5 & 6. The relative minus sign comes from swapping external product electrons between the two diagrams. Like Schwartz I suppress the spin indices on the single mode spinor fields u and


 FIG. 6. $\mathfrak{Y}$

v .

$$\mathfrak{Y} = -ie^2 \int d^4p \bar{u}(\vec{p}_3) \gamma^\mu u(\vec{p}_1) \frac{\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon} \bar{u}(\vec{p}_4) \gamma^\nu u(\vec{p}_2) \delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_4) \quad (71)$$

$$\mathfrak{Y} = -i\delta^{(4)}(\vec{P} - \vec{K}) e^2 \bar{u}(\vec{p}_3) \gamma^\mu u(\vec{p}_1) \frac{\eta_{\mu\nu}}{t} \bar{u}(\vec{p}_4) \gamma^\nu u(\vec{p}_2) \quad (72)$$

$$\eta_{\mu\nu} \gamma^\nu = \gamma_\mu \quad (73)$$

$$\mathfrak{Y} = -i\delta^{(4)}(\vec{P} - \vec{K}) e^2 \frac{[\bar{u}(\vec{p}_3) \gamma^\mu u(\vec{p}_1)] [\bar{u}(\vec{p}_4) \gamma_\mu u(\vec{p}_2)]}{t} \quad (74)$$

$$\mathfrak{Y} = -ie^2 \int d^4p \bar{u}(\vec{p}_4) \gamma^\mu u(\vec{p}_1) \frac{\eta_{\mu\nu}}{\vec{p}^2 + i\epsilon} \bar{u}(\vec{p}_3) \gamma^\nu u(\vec{p}_2) \delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3) \quad (75)$$

$$\mathfrak{Y} = -i\delta^{(4)}(\vec{P} - \vec{K}) e^2 \bar{u}(\vec{p}_4) \gamma^\mu u(\vec{p}_1) \frac{\eta_{\mu\nu}}{u} \bar{u}(\vec{p}_3) \gamma^\nu u(\vec{p}_2) \quad (76)$$

$$\mathfrak{Y} = -i\delta^{(4)}(\vec{P} - \vec{K}) e^2 \frac{[\bar{u}(\vec{p}_4) \gamma^\mu u(\vec{p}_1)] [\bar{u}(\vec{p}_3) \gamma_\mu u(\vec{p}_2)]}{u} \quad (77)$$

Adding both diagrams yields

$$\frac{\mathcal{M}}{e^2} = \frac{[\bar{u}(\vec{p}_4) \gamma^\mu u(\vec{p}_1)] [\bar{u}(\vec{p}_3) \gamma_\mu u(\vec{p}_2)]}{u} - \frac{[\bar{u}(\vec{p}_3) \gamma^\mu u(\vec{p}_1)] [\bar{u}(\vec{p}_4) \gamma_\mu u(\vec{p}_2)]}{t}. \quad (78)$$

My solution differs from Schwartz Eq. (13.37) by an overall minus sign, which does not affect observables. Why the difference? Because the overall sign does not matter Schwartz was less careful in applying the Feynman rules than I was. I want consistent rules for all cases and he was focused on getting the right answer.